

IT 214

Disaster Relief Management System

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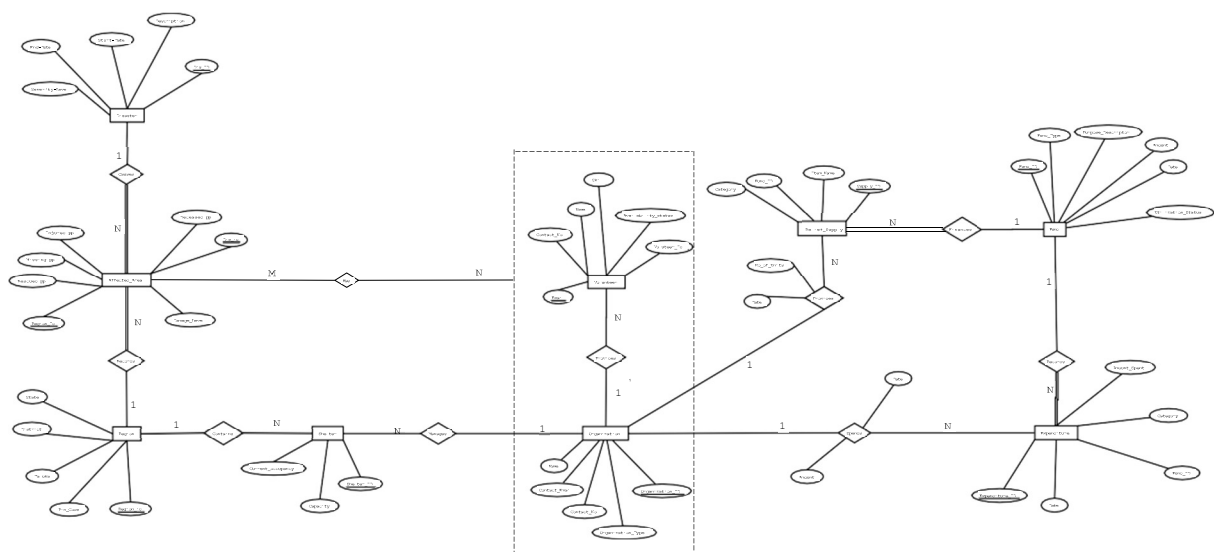
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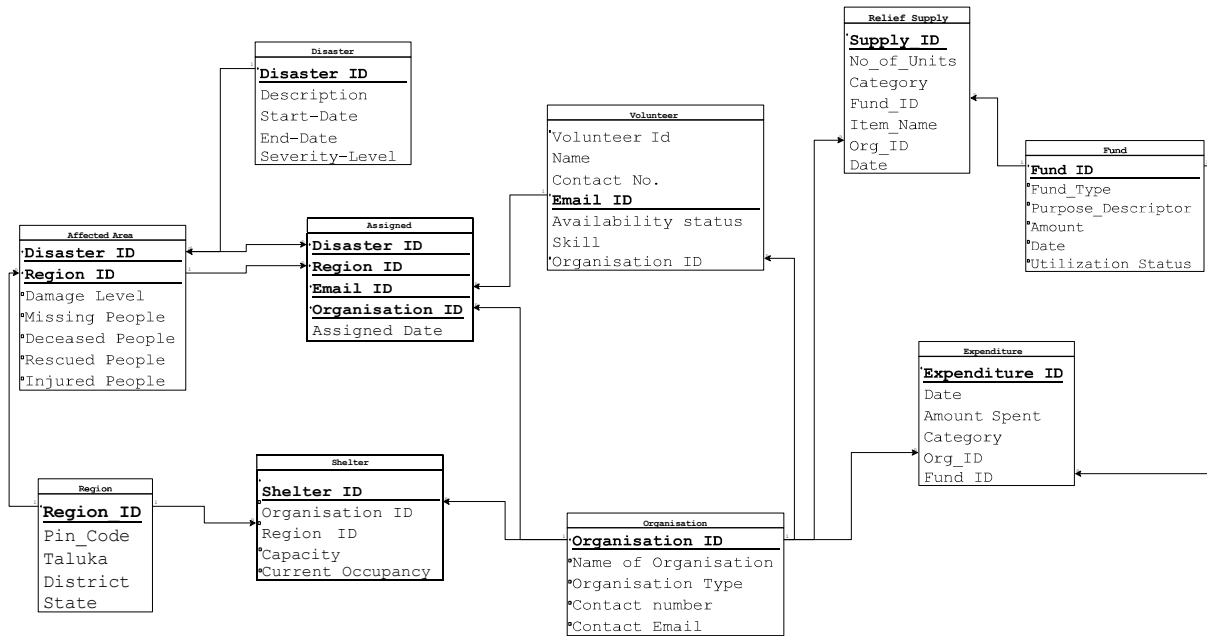
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1. Relational Schema



Functional Dependence Analysis

Part – 1 : List of Functional Dependence for Entities

1--- (Disaster)

FDs:

Dis_ID → Start_Date, End_Date, Severity_Level, Description, Damage_Level, Region_ID

Dis_ID → Deceased_ppl, Injured_ppl, Missing_ppl, Rescued_ppl

Notes:

Casualty figures recorded per disaster via the 'Records' relationship.

Region_ID is a FK → Region (1:N — each disaster belongs to one region).

2--- (Region)

FDs:

Region_ID → District, Taluka, Pin_Code, State, Affected_Area

3--- (Shelter)

FDs:

Shelter_ID → Capacity, Current_occupancy

Shelter_ID → Region_ID

Notes:

Each shelter is contained within one region (Contains: 1:N)

4--- (Relief_Supply)

FDs:

Supply_ID → Category, No_of_Units, Item_Name

Supply_ID → Fund_ID

Notes:

Fund_ID is FK → Fund (supply is financed by one fund).

5--- (Fund)

FDs:

Fund_ID → Fund_Type, Purpose_Descriptor, Amount, Date, Utilization_Status
Fund_ID → Organization_ID

Notes:

Organization_ID is FK → Organization (fund is financed by one organization).

Fund_ID → Name, Organization_Type, Contact_No, Contact_Email *TRANSITIVE via Organization_ID+

6--- (Expenditure)

FDs:

Expenditure_ID → Category, Amount_Spent, Date
Expenditure_ID → Fund_ID

Notes:

Fund_ID is FK → Fund (expenditure spends from one fund).

Expenditure_ID → Fund_Type, Amount, Utilization_Status *TRANSITIVE via Fund_ID+

7--- (Organization)

FDs:

Organization_ID → Name, Organization_Type, Contact_No, Contact_Email

8--- (Volunteer)

FDs:

Email_ID → Skill, Name, Contact_No, Availability_status

Notes:

Volunteer is now on the M side of TWO M:N relationships:

(a) Volunteer — Has — Disaster (M:N) → junction table created

(b) Volunteer — Provides — Organization (M:N) → junction table created

Therefore Volunteer does NOT carry Disaster_ID or Organization_ID as FK.

9--- (NEW junction table – M:N ‘HAS’ relation)

FDs:

(Email_ID, Dis_ID) → Date, Amount

Notes:

Composite PK: (Email_ID, Dis_ID).

Date and Amount are relationship attributes on the 'Has' relationship.

Both columns are also FKs: Email_ID → Volunteer, Dis_ID → Disaster.

10--- (NEW junction table – M:N ‘PROVIDES’ relation)

FDs:

(Email_ID, Organization_ID) → [no extra attributes]

Transitive FDs present in Full Set:

Transitive FDs	Derivation Chain
Dis_ID → District, Taluka, Pin_Code, State, Affected_Area	Dis_ID → Region_ID → Region attributes
Fund_ID → Name, Org_Type, Contact_No, Contact_Email	Fund_ID → Organization_ID → Organization attributes
Expenditure_ID → Fund_Type, Amount, Utilization_Status	Expenditure_ID → Fund_ID → Fund attributes
Supply_ID → Fund_Type, Amount, Utilization_Status	Supply_ID → Fund_ID → Fund attributes

Part – 2 : Minimal FD Set for the Entities

1--- (Disaster)

Minimal FD Set:

Dis_ID → Start_Date
Dis_ID → End_Date
Dis_ID → Severity_Level
Dis_ID → Description
Dis_ID → Damage_Level
Dis_ID → Region_ID
Dis_ID → Deceased_ppl
Dis_ID → Injured_ppl
Dis_ID → Missing_ppl
Dis_ID → Rescued_ppl

2--- (Region)

Minimal FD Set:

Region_ID → District
Region_ID → Taluka
Region_ID → Pin_Code
Region_ID → State
Region_ID → Affected_Area

3--- (Shelter)

Minimal FD Set:

Shelter_ID → Capacity
Shelter_ID → Current_occupancy
Shelter_ID → Region_ID

4--- (Relief_Supply)

Minimal FD Set:

Supply_ID $\rightarrow$ Category
Supply_ID $\rightarrow$ No_of_Units
Supply_ID $\rightarrow$ Item_Name
Supply_ID $\rightarrow$ Fund_ID

5--- (Fund)

Minimal FD Set:

Fund_ID $\rightarrow$ Fund_Type
Fund_ID $\rightarrow$ Purpose_Descriptor
Fund_ID $\rightarrow$ Amount
Fund_ID $\rightarrow$ Date
Fund_ID $\rightarrow$ Utilization_Status
Fund_ID $\rightarrow$ Organization_ID

6--- (Expenditure)

Minimal FD Set:

Expenditure_ID $\rightarrow$ Category
Expenditure_ID $\rightarrow$ Amount_Spent
Expenditure_ID $\rightarrow$ Date
Expenditure_ID $\rightarrow$ Fund_ID

7--- (Organization)

Minimal FD Set:

Organization_ID $\rightarrow$ Name
Organization_ID $\rightarrow$ Organization_Type
Organization_ID $\rightarrow$ Contact_No
Organization_ID $\rightarrow$ Contact_Email

8--- (Volunteer)

Minimal FD Set:

Email_ID $\rightarrow$ Skill

Email_ID $\rightarrow$ Name

Email_ID $\rightarrow$ Contact_No

Email_ID $\rightarrow$ Availability_status

9--- (Volunteer_Disaster – M:N junction with ‘HAS’ relation)

Minimal FD Set:

(Email_ID, Dis_ID) $\rightarrow$ Date

(Email_ID, Dis_ID) $\rightarrow$ Amount

10--- (Volunteer_Organization – M:N junction with ‘PROVIDES’ relation)

Minimal FD Set:

No non-key attributes — PK (Email_ID, Organization_ID) is the only FD

Part – 3 : FDs removed for the Minimal FD Set

Removed FDs	Type	Available Derivable path
Dis_ID → District, Taluka, Pin_Code, State, Affected Area	Transitive	Dis_ID → Region_ID → Region attributes
Fund_ID → Name, Organization_Type, Contact No, Contact Email	Transitive	Fund_ID → Organization_ID → Org attributes
Expenditure_ID → Fund_Type, Amount, Purpose_Descriptor, Utilization_Status	Transitive	Expenditure_ID → Fund_ID → Fund attributes
Supply_ID → Fund_Type, Amount, Purpose_Descriptor, Utilization_Status	Transitive	Supply_ID → Fund_ID → Fund attributes
Volunteer.Disaster_ID (as FK) Volunteer.Organization_ID (as FK)	Structural Change	M:N relationships replaced with junction tables; FKs moved to junction tables

BCNF Proof For Each Relation

1--- (Disaster)

Schema:

Disaster(Disaster_ID, Description, Start_Date, End_Date, Severity_Level, Region_ID)

FDs:

Disaster_ID → Description, Start_Date, End_Date, Severity_Level, Region_ID

BCNF Verification:

Only FD present: Disaster_ID → {all other attributes}

Disaster_ID is the Primary Key, hence a Superkey

Every determinant (Disaster_ID) is a Superkey

No partial dependency — single-attribute PK

No transitive dependency — Region_ID is a FK, not a determinant here

Hence, Relation is in BCNF.

2--- (Region)

Schema:

Region(Region_ID, Pin_Code, Taluka, District, State)

FDs:

Region_ID $\rightarrow$ Pin_Code, Taluka, District, State

BCNF Verification:

Only FD present: Region_ID $\rightarrow$ {Pin_Code, Taluka, District, State}

Region_ID is the Primary Key, hence a Superkey

Every determinant is a Superkey

No partial or transitive dependencies exist

Hence, Relation is in BCNF.

3--- (Shelter)

Schema:

Shelter(Shelter_ID, Capacity, Current_Occupancy, Region_ID)

FDs:

Shelter_ID $\rightarrow$ Capacity, Current_Occupancy, Region_ID

BCNF Verification:

Only FD present: $\text{Shelter_ID} \rightarrow \{\text{Capacity}, \text{Current_Occupancy}, \text{Region_ID}\}$

Shelter_ID is the Primary Key, hence a Superkey

Region_ID is a FK referencing Region — it does not determine any attribute here

No partial or transitive dependencies exist

Hence, Relation is in BCNF.

4--- (Organization)

Schema:

Organization(Organization_ID , Name, Contact_No, Contact_Email, Organization_Type)

FDs:

$\text{Organization_ID} \rightarrow \text{Name}, \text{Contact_No}, \text{Contact_Email}, \text{Organization_Type}$

BCNF Verification:

Only FD present: $\text{Organization_ID} \rightarrow \{\text{all attributes}\}$

Organization_ID is the Primary Key, hence a Superkey

No non-prime attribute determines another non-prime attribute

Hence, Relation is in BCNF.

5--- (Volunteer)

Schema:

Volunteer(Volunteer_ID , Name, Skill, Contact_No, Email, Availability_Status, Disaster_ID, Organization_ID)

FDs:

Email_ID $\rightarrow$ Name, Skill, Contact_No, Availability_Status, Disaster_ID, Organization_ID

BCNF Verification:

Only FD present: Email_ID $\rightarrow$ {all other attributes}

Email_ID is the Primary Key, hence a Superkey

Disaster_ID and Organization_ID are FKs — they do not determine any attribute

No partial or transitive dependencies exist

Hence, Relation is in BCNF.

6--- (Fund)

Schema:

Fund(Fund_ID, Fund_Type, Purpose_Descriptor, Amount, Date, Utilization_Status)

FDs:

Fund_ID $\rightarrow$ Fund_Type, Purpose_Descriptor, Amount, Date, Utilization_Status

BCNF Verification:

Only FD present: Fund_ID $\rightarrow$ {all attributes}

Fund_ID is the Primary Key, hence a Superkey

No non-key attribute determines another attribute

Hence, Relation is in BCNF.

7--- (Relief_Supply)

Schema:

Relief_Supply(Supply_ID, No_of_Units, Category, Item_Name, Fund_ID, Org_ID)

FDs:

Supply_ID $\rightarrow$ No_of_Units, Category, Item_Name, Fund_ID, Org_ID

BCNF Verification:

Only FD present: Supply_ID $\rightarrow$ {all other attributes}

Supply_ID is the Primary Key, hence a Superkey

Fund_ID and Org_ID are FKs — they do not determine any other attribute here

No partial or transitive dependencies exist

Hence, Relation is in BCNF.

8--- (Affected_Area)

Schema:

Affected_Area(Disaster_ID, Deceased_Ppl, Injured_Ppl, Missing_Ppl, Rescued_Ppl)

FDs:

Disaster_ID $\rightarrow$ Deceased_Ppl, Injured_Ppl, Missing_Ppl, Rescued_Ppl

BCNF Verification:

`Disaster_ID` is the Primary Key (FK referencing Disaster), hence a Superkey

Casualty attributes depend solely on `Disaster_ID`

No non-key attribute determines another

Hence, Relation is in BCNF.

DDL Script for the below Relational Schema

-- DISASTER

```
CREATE TABLE Disaster (  
    Disaster_ID INT PRIMARY KEY,  
    Description VARCHAR(255),  
    Start_Date DATE,  
    End_Date DATE,  
    Severity_Level VARCHAR(50)  
);
```

-- REGION

```
CREATE TABLE Region (  
    Region_ID INT PRIMARY KEY,  
    Pin_Code VARCHAR(10),  
    Taluka VARCHAR(100),  
    District VARCHAR(100),  
    State VARCHAR(100)  
);
```

-- ORGANISATION

```
CREATE TABLE Organisation (  
    Organisation_ID INT PRIMARY KEY,  
    Name VARCHAR(255),  
    Organisation_Type VARCHAR(100),  
    Contact_Number VARCHAR(15),  
    Contact_Email VARCHAR(100)  
);
```

-- FUND

```
CREATE TABLE Fund (  
    Fund_ID INT PRIMARY KEY,  
    Fund_Type VARCHAR(100),  
    Purpose_Descriptor VARCHAR(255),  
    Amount DECIMAL(12,2),  
    Date DATE,  
    Utilization_Status VARCHAR(50)  
);
```

-- RELIEF_SUPPLY

```
CREATE TABLE Relief_Supply (  
    Supply_ID INT PRIMARY KEY,  
    No_of_Units INT,  
    Category VARCHAR(100),
```



```
Item_Name VARCHAR(255),

Date DATE,

Fund_ID INT,

Org_ID INT,

FOREIGN KEY (Fund_ID) REFERENCES Fund(Fund_ID),

FOREIGN KEY (Org_ID) REFERENCES Organisation(Organisation_ID)

);
```

-- EXPENDITURE

```
CREATE TABLE Expenditure (

Expenditure_ID INT PRIMARY KEY,

Date DATE,

Amount_Spent DECIMAL(12,2),

Category VARCHAR(100),

Org_ID INT,

Fund_ID INT,

FOREIGN KEY (Org_ID) REFERENCES Organisation(Organisation_ID),

FOREIGN KEY (Fund_ID) REFERENCES Fund(Fund_ID)

);
```

-- VOLUNTEER

```
CREATE TABLE Volunteer (

Volunteer_ID INT ,

Name VARCHAR(255),

Contact_No VARCHAR(15),

Email_ID VARCHAR(100) PRIMARY KEY,

Availability_Status VARCHAR(50),
```

```
Skill VARCHAR(100),  
  
Organisation_ID INT,  
  
FOREIGN KEY (Organisation_ID) REFERENCES Organisation(Organisation_ID)  
  
);
```

```
-- SHELTER
```

```
CREATE TABLE Shelter (  
  
    Shelter_ID INT PRIMARY KEY,  
  
    Organisation_ID INT,  
  
    Region_ID INT,  
  
    Capacity INT,  
  
    Current_Occupancy INT,  
  
    FOREIGN KEY (Organisation_ID) REFERENCES Organisation(Organisation_ID),  
  
    FOREIGN KEY (Region_ID) REFERENCES Region(Region_ID)  
  
);
```

```
-- AFFECTED_AREA
```

```
CREATE TABLE Affected_Area (  
  
    Disaster_ID INT,  
  
    Region_ID INT,  
  
    Damage_Level VARCHAR(50),  
  
    Missing_People INT,  
  
    Deceased_People INT,  
  
    Rescued_People INT,  
  
    Injured_People INT,  
  
    PRIMARY KEY (Disaster_ID, Region_ID),  
  
    FOREIGN KEY (Disaster_ID) REFERENCES Disaster(Disaster_ID),
```

```
FOREIGN KEY (Region_ID) REFERENCES Region(Region_ID)

);

-- ASSIGNED (Relationship table)

CREATE TABLE Assigned (

    Disaster_ID INT,

    Region_ID INT,

    Organisation_ID INT,

    Email_ID VARCHAR(100),

    Assigned_Date DATE,

    PRIMARY KEY (Disaster_ID, Region_ID, Organisation_ID),

    FOREIGN KEY (Disaster_ID) REFERENCES Disaster(Disaster_ID),

    FOREIGN KEY (Region_ID) REFERENCES Region(Region_ID),

    FOREIGN KEY (Organisation_ID) REFERENCES Organisation(Organisation_ID),

    FOREIGN KEY (Email_ID) REFERENCES Volunteer(Email_ID)

);
```